

Synthesis of Urea As a Nitrogenous Fertilizer

Design Analysis of an Integrated Ammonia-Urea Production Plant Utilizing the Haber-Bosch and Stamicarbon CO₂ Stripping Processes

Group 2

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CHE 396 Senior Design

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Introduction

- Population is increasing
- Global agriculture depends on synthetic fertilizers
- Ammonia and urea are the most common N based
- Urea's low cost per unit N is ideal
- High potential for sustainability



"Farmer Is Pouring Chemical Fertilizer by Hand over Green Background." *iStock*, 4 Oct. 2012, www.istockphoto.com/photo/farmer-pouring-chemical-fertilizer-in-the-field-gm153685368-21646496. Accessed 17 Sept. 2025.

Plant Design

- Based on the QAFCO plant, Qatar
- 3.8 million MT ammonia and 5.6 million MT urea annually
- Ras Laffan Industrial City handles NG
- Optimal location for trade: India, China, SE Asia, Africa, and Europe



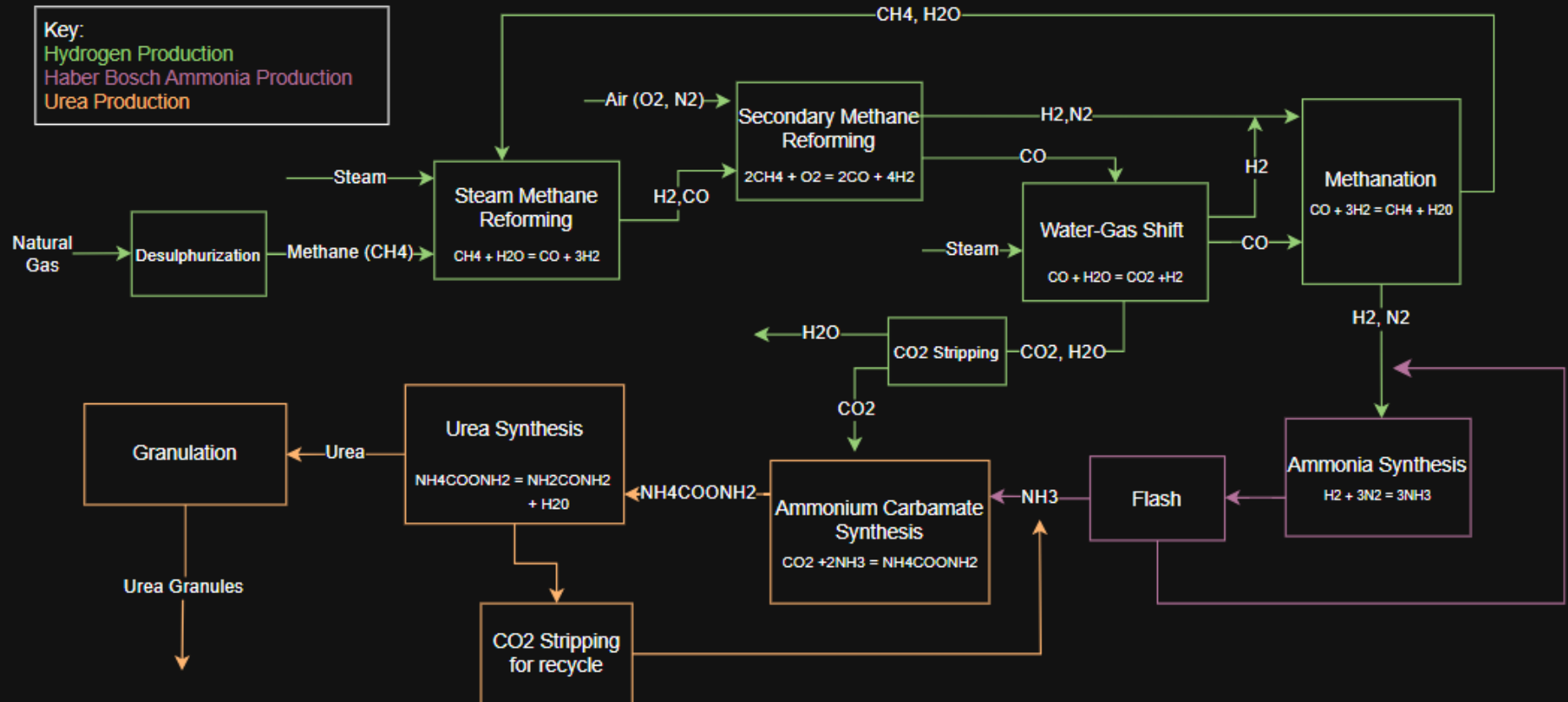
"Urea Plant Mesaieed QAFCO." *Mesit.com*, 2025, www.mesit.com/en/projects/3/urea-plant-mesaieed-qafco.html. Accessed 17 Sept. 2025.

Design Process

- Hydrogen Production
 - SMR
 - WGS
- Ammonia Production
 - Haber-Bosch Process
- Urea Production
 - Stamicarbon CO₂ Stripping process

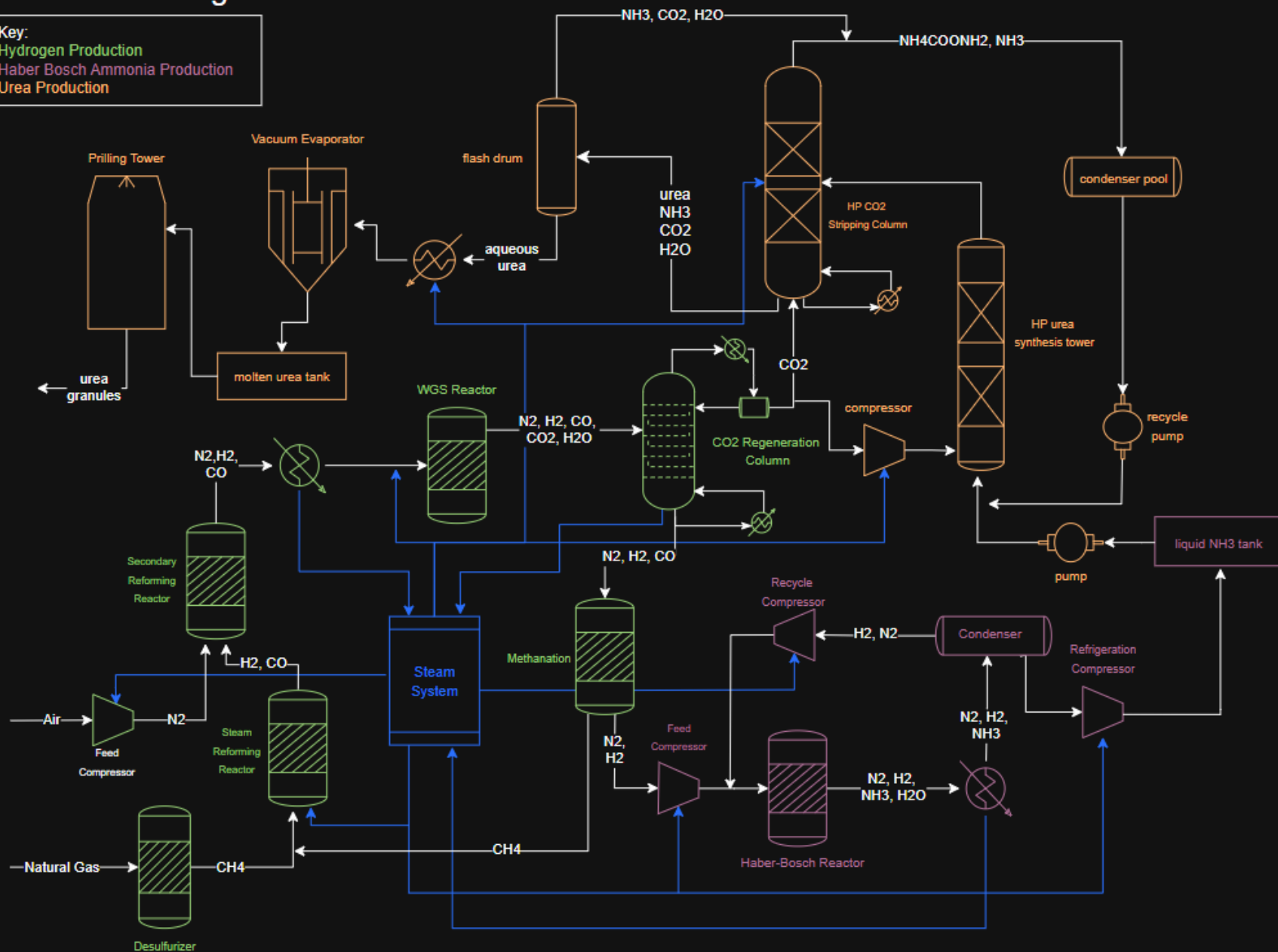
Block Flow Diagram

Key:
 Hydrogen Production
 Haber Bosch Ammonia Production
 Urea Production



Process Flow Diagram

Key:
 Hydrogen Production
 Haber Bosch Ammonia Production
 Urea Production

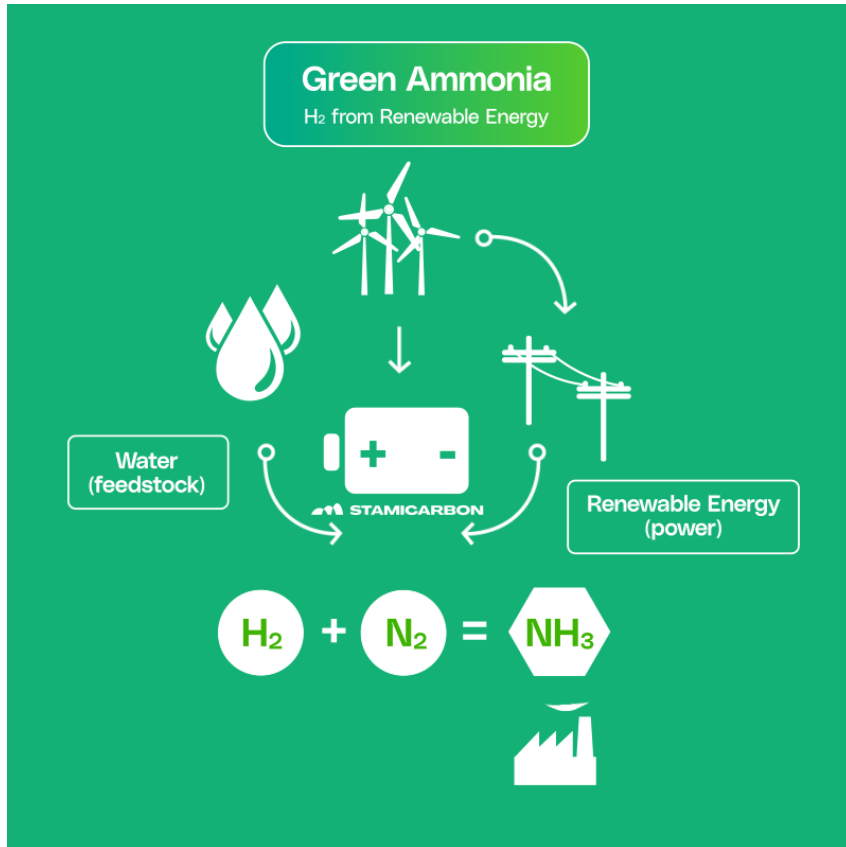


Green Ammonia (Counter process)



- Conventional ammonia 1–2% global CO₂ emissions
- Green ammonia uses renewable-powered electrolysis near-zero CO₂
- Key roles: Decarbonizing fertilizers (low-carbon nitrogen products)
- Zero-carbon fuel for shipping & power

Process Overview



1. Hydrogen from water electrolysis
2. Nitrogen from cryogenic air separation
3. $\text{H}_2 + \text{N}_2 \rightarrow \text{Ammonia}$ via Haber–Bosch
4. Ammonia + captured $\text{CO}_2 \rightarrow \text{Urea}$ (net-zero)

Challenges & Opportunities

- Challenges:
 - High electricity demand
 - Costly electrolyzers & durability issues
- Opportunities:
 - Net-zero fertilizers
 - Sustainable fuel for global energy transition

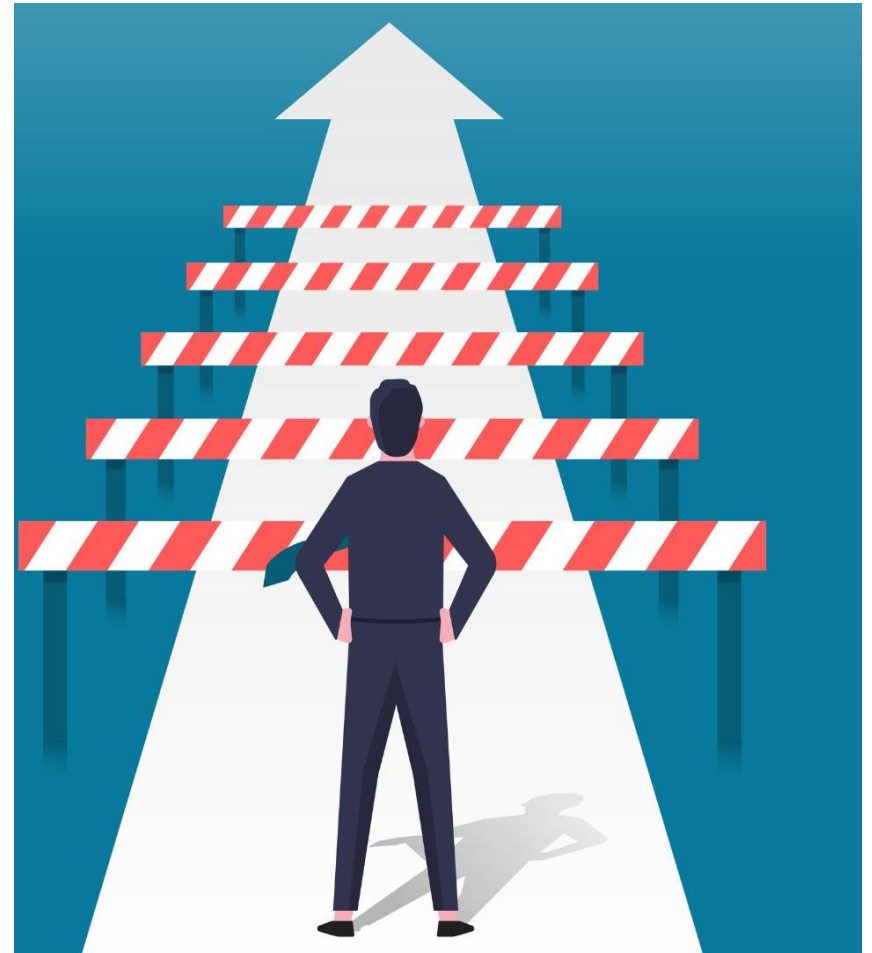


Market Analysis

- **Market value:** USD 128.9 billion in 2023; projected to reach USD 160–160.7 billion by 2032 (Fortune Business Insights, 2023).
- **Global production & consumption:** 180 million tons in 2024, expected to reach 276 million tons by 2035 at 3.9–4.0% CAGR (ChemAnalyst, 2024).
- **Asia-Pacific dominates:** China, India, and other large agricultural nations control over 50% of global market share (ChemAnalyst, 2024)

Market Analysis

- **Market challenges:**
- Price instability of natural gas.
- Growing demand for **green urea production** to reduce CO₂ emissions.

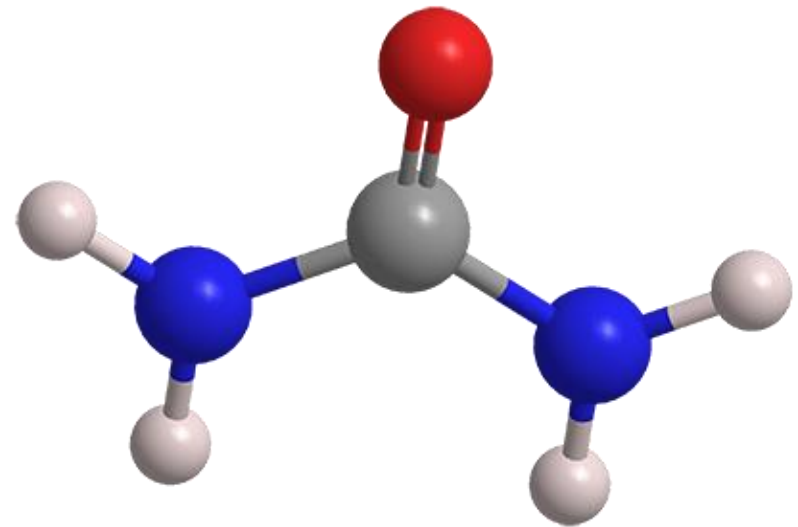


Market Analysis

- **Raw Materials / Inputs Costs :**
- **Natural Gas Costs:**
 - Requirement: **16 MMBtu per ton of urea**
 - Price range: **\$2.89–\$3.10 per MMBtu** (EIA, 2025; FRED, 2025).
 - Cost impact: **\$47–\$51 per ton of urea.**
- **Water Costs:**
 - Requirement: **5–13 m³ per ton of urea**
 - Price: **\$0.74 per m³** (City of Bryan, 2021).
 - Cost impact: **\$4–\$10 per ton of urea.**
- **Air & CO₂ Costs: Free**

Market Analysis

- **Total Production Cost:**
- **Raw materials:** \$51–\$61 per ton of urea.
- **Annual facility output:** 5.6 million tons of urea.
- **Yearly raw material expenses:** \$285–\$340 million.

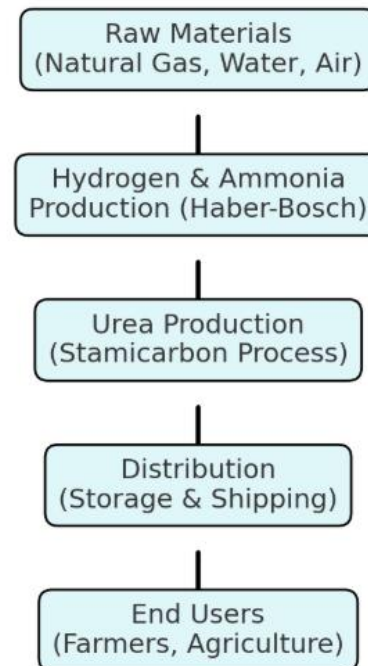


Market Analysis

Supply Chain

Supply Chain (Flow of Materials)

- Raw materials
- Ammonia production (Haber-Bosch)
- Urea production (Stamicarbon)
- Final processing
- Distribution channels
- End users

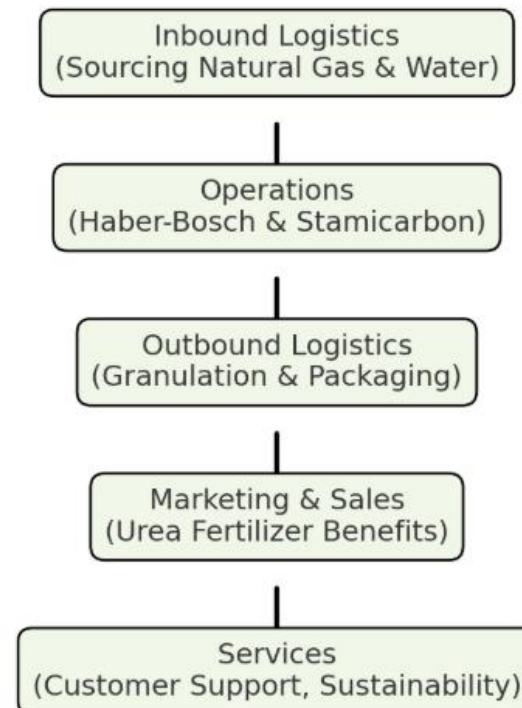


Market Analysis

Value Chain

- Inbound Logistics
- Operations (Haber-Bosch & Stamicarbon)
- Outbound Logistics
- Marketing & Sales
- After-Sales Services

Value Chain (Value Creation)



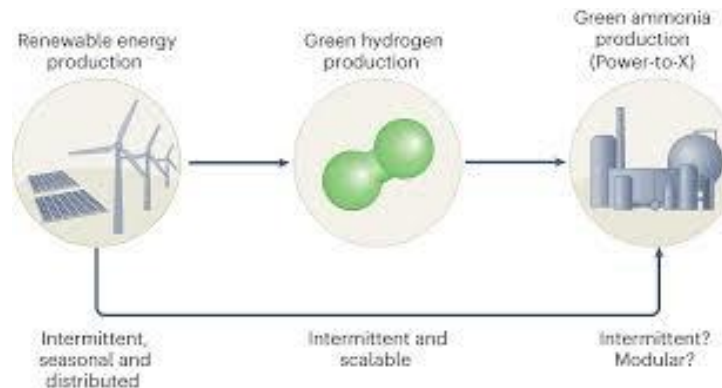
Market Analysis (Green Ammonia)

Market Value

- **Estimates:** \$44-\$445.7 million (2023-2024) and projected to reach \$44-73 billion in 2030.
- **Growth:** Rapid growth of 65% to 116% from 2024 to 2030.

Use

- Green Ammonia is mainly used in **fertilizers**.
- Significant demand/traction as a carbon-free fuel for **energy storage** and **maritime industry**.



Market Analysis (Green Ammonia)

Market Drivers

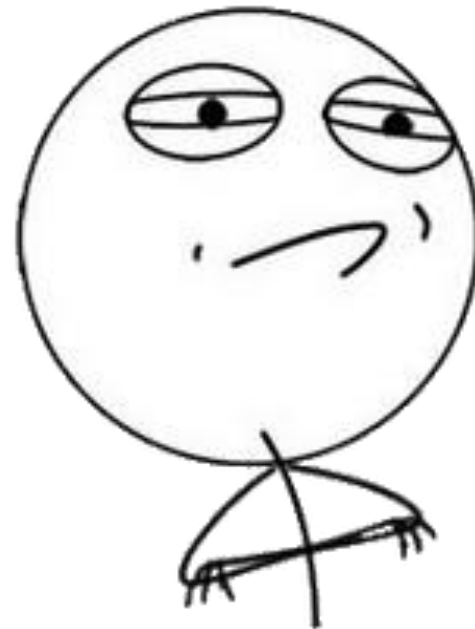
- **Decarbonization:** Governments and industries prioritize reducing emissions for net-zero goals.
- **Incentives:** EU and US legislation offers incentives to producers



Market Analysis (Green Ammonia)

Challenges

- **1.5x** more expensive than traditional ammonia (from fossil fuels).
- Expensive capital cost due to renewable energy infrastructure.
- Storing green ammonia is **EXPENSIVE**.
- Uncertainty with government policies (delays, changes, etc.)



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